

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3445U10-1



S23-3445U10-1

FRIDAY, 16 JUNE 2023 – MORNING**APPLIED SCIENCE (Double Award)****UNIT 1: Energy, Resources and the Environment****FOUNDATION TIER**

1 hour 30 minutes

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **3(b)** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A Periodic Table is printed on page 20.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	14	
3.	9	
4.	12	
5.	11	
6.	8	
7.	11	
Total	75	



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Answer **all** questions.

1. Ffion is investigating a neutralisation reaction between sulfuric acid and sodium hydroxide.

(a) Use the correct words from the box to complete the **word** equation for this reaction below. [1]

sodium chloride sodium nitrate sodium sulfate

sulfuric acid + sodium hydroxide → + water

(b) In the experiment she monitors the change of pH using both a pH meter and an indicator.

The chart below shows the colour of phenolphthalein indicator at different pH.

pH scale	1	2	3	4	5	6	7	8	9	10	11	12	13	14
phenolphthalein	colourless								pale pink	pink				

(i) The table below shows Ffion's results. Complete the table. [2]

Volume of sodium hydroxide added (cm ³)	pH meter reading	Phenolphthalein indicator colour
0	1.0	colourless
5	1.0	colourless
10	1.1	colourless
15	1.2	colourless
20	2.0	
25	9.0	pale pink
30	13.0	
35	14.0	pink
40	14.0	pink



- (ii) Use the information in the table to tick (✓) the boxes next to the **three** correct statements. [3]

Before any sodium hydroxide is added to the acid the phenolphthalein turns green.

☐

The pH is lowest at the start of the experiment.

☐

When 10 cm³ of sodium hydroxide has been added the solution is neutral.

☐

As sodium hydroxide is added the pH rises.

☐

The pH is highest at the end of the experiment.

☐

When 40 cm³ of sodium hydroxide has been added the solution is acidic.

☐

- (c) Sodium hydroxide (NaOH) contains the three elements sodium, hydrogen and oxygen.

- (i) Select the correct numbers from the box to complete the table below. [3]

1	2	8	11	12	16
---	---	---	----	----	----

Element	Number of protons	Number of neutrons	Number of electrons	Mass number
sodium	11	12		23
oxygen	8		8	16
hydrogen	1	0	1	

- (ii) Sodium metal is found in Group 1 of the Periodic Table. Use the Periodic Table on page 20 to name **one** other metal found in this group. [1]

.....



2. The UK has been generating more energy using wind power over recent years.



- (a) Circle **one** renewable energy source below.

[1]

coal

gas

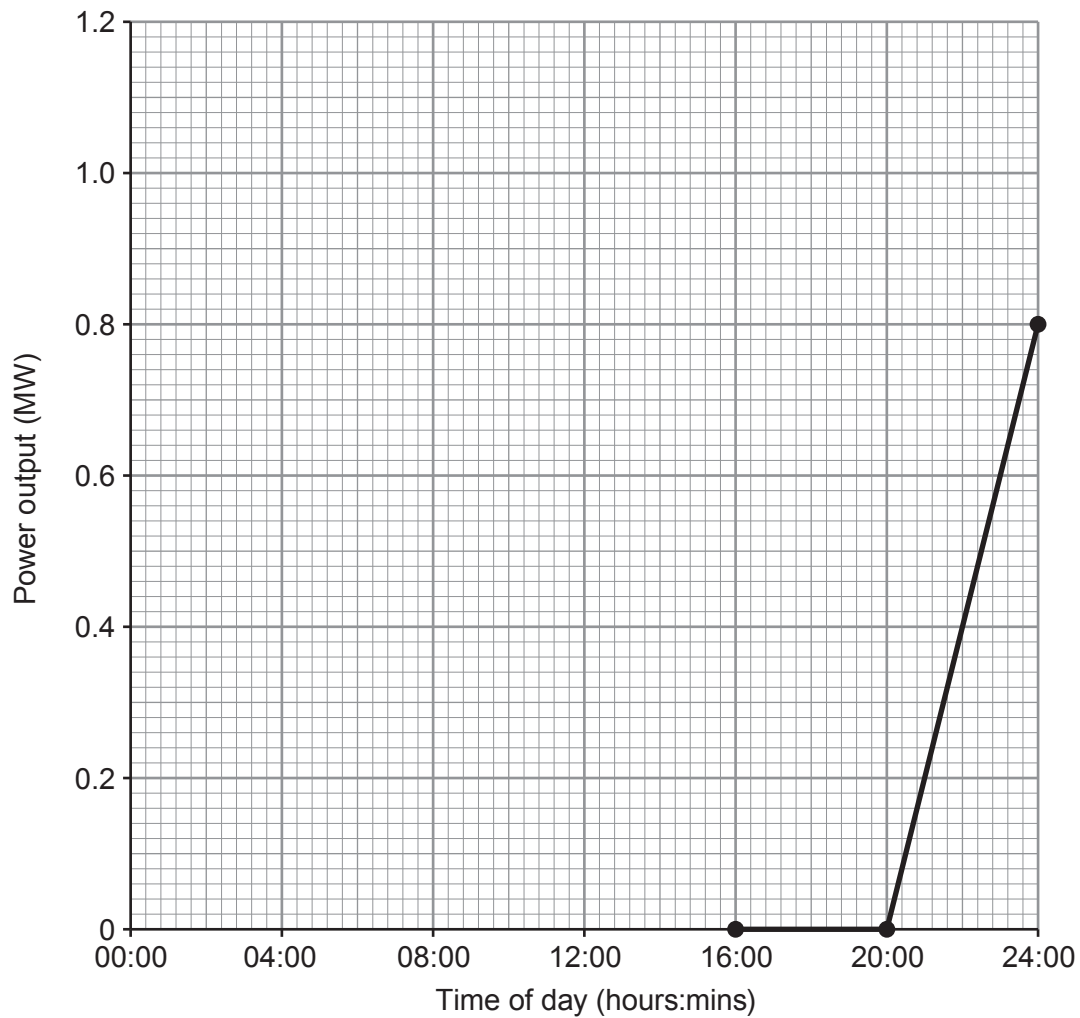
solar

- (b) The information below shows the power output of a wind turbine in one day.

Time of day (hours:mins)	Power output of wind turbine (MW)
00:00	0.6
04:00	1.0
08:00	1.2
12:00	0.8
16:00	0.0
20:00	0.0
24:00	0.8



- (i) Use the data from the table to complete the graph on the grid below and join the points with a ruler. The last three points have been plotted for you. [3]



- (ii) State why the power generated by this wind turbine is not constant during this day. [1]

.....

.....



(iii) Use the information to answer the following questions.

I. State the maximum power produced by the wind turbine during this day. [1]

maximum power = MW

II. State the time of day that the wind speed was greatest. [1]

time =

III. Determine between which hours the wind turbine did not generate any electricity. [1]

from: to:

(iv) David suggested that the wind turbine generated a power greater than 0.6 MW for 16 hours.

Add a horizontal line to your graph and use it to explain whether David was correct. [4]

.....

.....

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.....

(c) Last year this wind turbine had a mean power output of 0.6 MW. A typical nuclear power station has a continuous power output of 3600 MW.

Calculate how many wind turbines would be needed to generate the same power output as a nuclear power station. [2]

number of wind turbines =

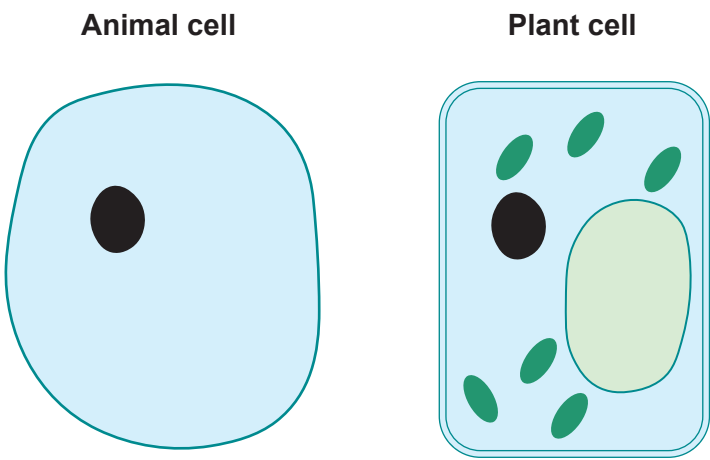


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3. Clara is studying the structure and function of cells. The images below show animal and plant cells.



(a) Complete the table below to show which parts are present in animal cells. [3]

Part	Plant cells	Animal cells (✓ or ×)
cell wall	✓	×
cytoplasm	✓	✓
chloroplast	✓	×
nucleus	✓	
cell membrane	✓	
vacuole	✓	



(b) Describe the functions of the parts of a **plant cell** listed in the table.

[6 QER]

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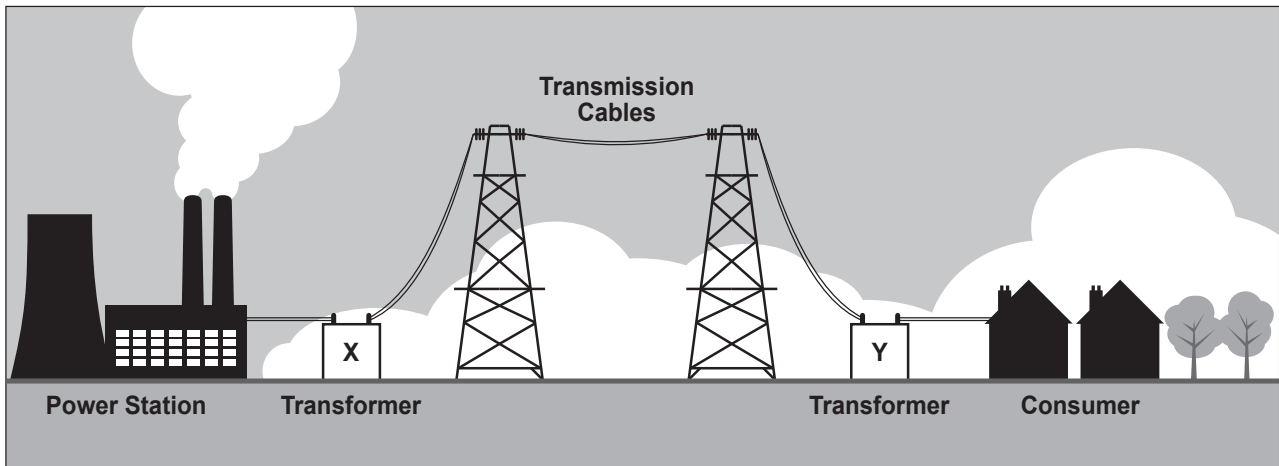
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4. The diagram below shows part of the National Grid.



- (a) Complete the sentences about the National Grid by selecting the correct words from the box. Each word can be used once, more than once or not at all. [4]

faster reliable increases generates safely decreases

The National Grid makes sure that the electricity supply to consumers is

.....

Transformer X the voltage and
the current.

Transformer Y changes the voltage so that consumers can use it
..... in their homes.



- (b) Five electrical supply companies (**A**, **B**, **C**, **D** and **E**) each produce the same amount of electrical power.

The table below shows the percentage of electrical power output from each energy source by each company.

Percentage of electrical power output from each energy source (%)					
Energy source	Company A	Company B	Company C	Company D	Company E
coal	4.8	5.8	7.1	10.8	10.1
natural gas	63.0	55.5	9.6	63.0	46.0
nuclear	5.2	20.6	71.2	23.3	13.9
renewable	27.0	18.1	12.1	2.9	30.0

Use the table to answer the following questions.

- (i) State which energy company produces the most electrical power from coal. [1]

.....

- (ii) State which company produces the most nuclear waste. [1]

.....

- (iii) Calculate the **mean** percentage of electrical power output generated from **coal** by the five companies (**A**, **B**, **C**, **D** and **E**). [2]

mean percentage =

- (c) The carbon footprint of an energy supply customer can be an important factor when choosing a supply company.

- (i) State what is meant by the term 'carbon footprint'. [2]

.....

.....

- (ii) A customer decides against using Company **A** since they believe that they have the highest carbon footprint from generating electricity.

Use information from the table to explain whether the customer is correct. [2]

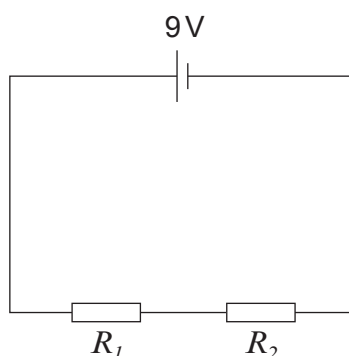
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5. Jamie is investigating an electrical circuit which has two **identical** resistors, R_1 and R_2 . The circuit that he built is shown below.



- (a) Complete the sentences by underlining the correct word(s) in the brackets.

- (i) In this series circuit the current in R_1 is (**the same as / higher than / lower than**) the current in R_2 . [1]
- (ii) In this series circuit the voltage across R_1 is (**the same as / higher than / lower than**) the voltage across R_2 . [1]

- (b) The voltage across R_1 is 4.5 V.

- (i) Each resistor was found to have a resistance of $2.25\ \Omega$.

I. Use the equation:

$$\text{current} = \frac{\text{voltage}}{\text{resistance}}$$

to calculate the current in R_1 . [2]

current = A

II. Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power developed in resistor R_1 . [2]

power developed in R_1 = W

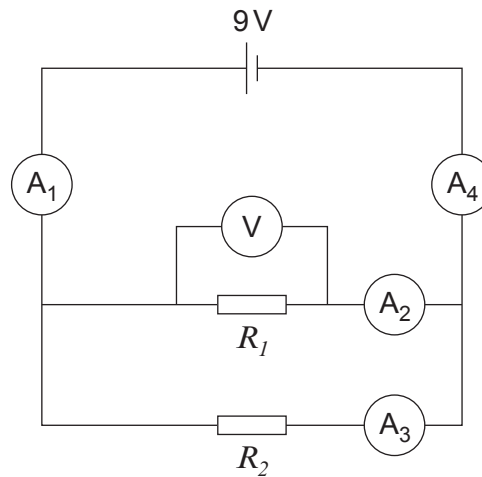


(ii) State the voltage across R_2 .

[1]

voltage across $R_2 = \dots\dots\dots$ V

(c) Jamie then built a parallel circuit using the same 2.25Ω resistors.



(i) State which ammeter (A_1 , A_2 , A_3 or A_4) measures the current through R_1 .

[1]

$\dots\dots\dots$

(ii) State the voltmeter reading across R_1 .

[1]

voltmeter reading = $\dots\dots\dots$ V

(iii) Ammeter A_2 reads 4 amps.

I. State the reading on ammeter A_3 .

[1]

ammeter reading = $\dots\dots\dots$ A

II. Calculate the reading shown on ammeter A_1 .

[1]

ammeter reading = $\dots\dots\dots$ A



6. Sioned reacted sulfuric acid with zinc oxide (ZnO).
She followed the method below to form zinc sulfate.

1. Pour 50 cm³ of 1.0 mol/dm³ sulfuric acid into a conical flask.
2. Heat the sulfuric acid to 50°C.
3. Add 5 g of zinc oxide to the warmed acid and stir.
4. Repeat step 3 until the zinc oxide is in excess.

- (a) The main hazard in this experiment is that 1.0 mol/dm³ sulfuric acid is an irritant.

- (i) Describe **one** risk associated with using sulfuric acid. [1]

.....
.....

- (ii) State a suitable control measure Sioned should carry out to reduce this risk. [1]

.....
.....

- (b) Circle the correct formula in each bracket to complete the symbol equation for this reaction. [2]



- (c) State **two** additional steps that Sioned needs to carry out to obtain pure zinc sulfate crystals. [2]

1.

.....

2.

.....

- (d) Robert has calculated the relative formula mass of zinc oxide to be 97.
Use the Periodic Table on page 20 to explain if he is correct. [2]

.....



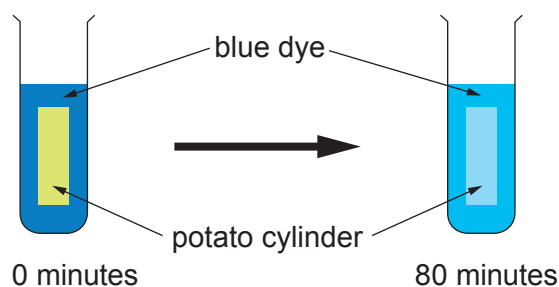
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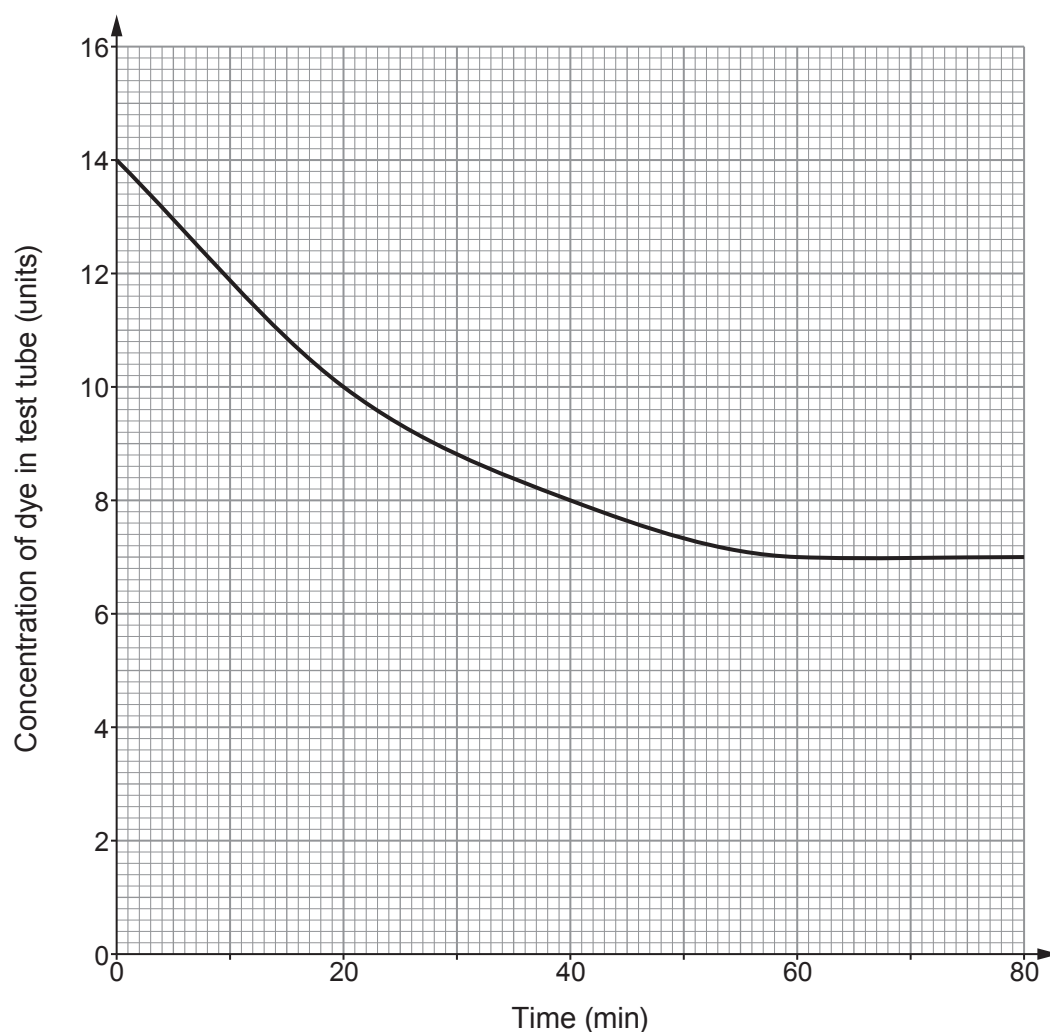


7. Several groups of students investigated the rate of diffusion of blue dye into potato by the method below.

1. Cut a cylinder of potato.
2. Place the cylinder in a test tube containing a solution of blue dye.
3. Put the test tube in a water bath at 20°C.
4. Measure the concentration of the dye after 20 minutes.
5. Continue to measure the concentration every 20 minutes up to 80 minutes.



The concentration of the dye left in the test tube after each 20-minute interval is shown on the graph below.



- (a) (i) State the dependent variable in this experiment. [1]

.....

- (ii) State **two** variables that should be controlled. [2]

1.

2.

- (b) (i) State how the concentration of dye left in the test tube changed during the first 20 minutes. [1]

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- (ii) Explain why this change occurred. [2]

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- (c) One student suggests that the rate of diffusion is constant over the whole 80 minutes. Explain whether you agree. [3]

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- (d) The students repeated the experiment in a water bath at 30°C. **Add another line** to the graph to show the expected results. [2]

END OF PAPER



[illegible]

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Key

A_r	Symbol	Name	Z
1	H	Hydrogen	1
2	He	Helium	2
3			
4	Li	Lithium	3
5	Be	Beryllium	4
6	B	Boron	5
7	C	Carbon	6
8	N	Nitrogen	7
9	O	Oxygen	8
10	F	Fluorine	9
11	Ne	Neon	10
12	Na	Sodium	11
13	Mg	Magnesium	12
14	Al	Aluminum	13
15	Si	Silicon	14
16	P	Phosphorus	15
17	S	Sulfur	16
18	Cl	Chlorine	17
19	Ar	Argon	18
20	K	Potassium	19
21	Ca	Calcium	20
22	Sc	Scandium	21
23	Ti	Titanium	22
24	V	Vanadium	23
25	Cr	Chromium	24
26	Mn	Manganese	25
27	Fe	Iron	26
28	Ni	Nickel	28
29	Cu	Copper	29
30	Zn	Zinc	30
31	Ga	Gallium	31
32	Ge	Germanium	32
33	As	Arsenic	33
34	Se	Selenium	34
35	Br	Bromine	35
36	Kr	Krypton	36
37	Rb	Rubidium	37
38	Sr	Strontium	38
39	Y	Yttrium	39
40	Zr	Zirconium	40
41	Nb	Niobium	41
42	Mo	Molybdenum	42
43	Tc	Technetium	43
44	Ru	Ruthenium	44
45	Rh	Rhodium	45
46	Pd	Palladium	46
47	Ag	Silver	47
48	Cd	Cadmium	48
49	In	Indium	49
50	Sn	Tin	50
51	Pb	Lead	82
52			
53	I	Iodine	53
54	Xe	Xenon	54
55	Ba	Barium	56
56	La	Lanthanum	57
57	Ce	Cerium	58
58	Pr	Praseodymium	59
59	Nd	Niodymium	60
60	Pm	Promethium	61
61	Sm	Samarium	62
62	Eu	Europium	63
63	Gd	Gadolinium	64
64	Tb	Terbium	65
65	Dy	Dysprosium	66
66	Ho	Holmium	67
67	Er	Erbium	68
68	Tm	Thulium	69
69	Yb	Ytterbium	70
70	Lu	Lutetium	71
71	Hf	Hafnium	72
72	Ta	Tantalum	73
73	W	Tungsten	74
74	Re	Rhenium	75
75	Os	Osmium	76
76	Ir	Iridium	77
77	Pt	Platinum	78
78	Au	Gold	79
79	Hg	Mercury	80
80	Tl	Thallium	81
81	Pb	Lead	82
82	Bi	Bismuth	83
83	Po	Polonium	84
84	At	Astatine	85
85	Rn	Radon	86
86	Fr	Francium	87
87	Ra	Radium	88
88	Ac	Actinium	89
89	Th	Thorium	90
90	Pa	Protactinium	91
91	U	Uranium	92
92	Np	Neptunium	93
93	Pu	Plutonium	94
94	Am	Americium	95
95	Cm	Curium	96
96	Bk	Berkelium	97
97	Cf	Californium	98
98	Es	Einsteinium	99
99	Fm	Fermium	100
100	Mendelevium		
101	Nobelium		
102	Lr	Lutetium	103
103	104		
104	105		
105	106		
106	107		
107	108		
108	109		

atomic number